Relationship between the Equipment table and the SRF Item table

How the two tables relate, when SRF Item data is written to Equipment, and where equipment updates are reflected.

1. The relationship between the tables

Four records are involved:

Record	NetSuite record	What it holds
SRF Item	`customrecord_twc_srf_itm`	One line of a Space Request: the equipment the customer is asking to install, remove or swap. This is the *request*.
Equipment Action	`customrecord_twc_eq_action`	The link between the two. One record per SRF Item (two for a swap). It points to the SRF, the SRF Item and the Equipment record. It also records the action type (Licence, Unlicence, Swap Licence, Swap Unlicence) and its status, and it's what a SAF attaches to.
Equipment	`customrecord_twc_equip`	The master record for a physical piece of equipment on a site. This is what Inventory, Site Info, SAF and SDS show.
SDS Equipment	`customrecord_twc_sds_item`	A snapshot of the equipment, taken when the SDS (licence) is generated.

The links work like this:

- The SRF Item has an **Equipment ID** field pointing to the Equipment record.
- The Equipment record has an **Associated Equipment Action** field pointing back to the action that created it.
- Each Equipment Action points to both the SRF Item and the Equipment record.

Each request type maps to equipment as follows:

- **Install:** one *Licence* action. A **new** Equipment record is created from the SRF Item.
- **Remove:** one *Unlicence* action, pointing to the **existing** Equipment record. When the line is added, the SRF Item's values are copied *from* that equipment.
- **Swap:** two actions.
  - An *Unlicence* action on the existing equipment.
  - A *Licence* action that creates a new Equipment record.
  - The SRF Item's Equipment ID stays on the equipment being swapped out.

2. When SRF Item contents are written to an Equipment record

- **While the SRF is in Draft:** only the SRF Item and Equipment Action records exist. No equipment is created or changed yet.
- **When the SRF is submitted:** the system copies the SRF Item into the Equipment record. For Install / Swap-in lines it creates the record. The fields copied are:
  - site and customer
  - class, type and structure
  - equipment library entry
  - make, model and description
  - length, width, depth and weight
  - height on tower, azimuth and B-End
  - customer reference, inventory flag, feeder type and voltage type
  - modified antenna location and GIE location
  - the parent TME (for ATME and feeders)

  The new equipment is created with Install Status **Not Installed** and Licence Status **Requested to Licence**. Remove lines only get their Licence Status set to **Requested to Unlicence**.
- **Each time the SRF is saved again** in Submitted, Under Review, Feedback Issued or Resubmitted, and when it's re-submitted after feedback, the same copy runs again. **The SRF Item overwrites those Equipment fields every time.**

- **If an SRF line is deleted while it can still be edited:** its Equipment Action is deleted. For an Install line, the Equipment record that was created for it is deleted as well.
- **After the SRF is approved,** no more descriptive data is copied. Only status fields are updated (see below).

The copy only goes one way, from SRF Item to Equipment. **Changes made on an Equipment record are never written back to the SRF Item.**

3. Where updates to a piece of equipment are reflected, and when

Licence Status (on the Equipment record) is driven by the SRF workflow

SRF status	Install / swap-in equipment	Remove / swap-out equipment
Submitted / Resubmitted	Requested to Licence	Requested to Unlicence
SRF Approved	Proposed to Licence	Proposed to Unlicence
Licence Requested	Proposed to Licence – Licence Requested	Proposed to Unlicence – Licence Requested
Licence Issued	Proposed to Licence – Licence Issued	Proposed to Unlicence – Licence Issued
Licence Executed	**Licenced** (plus Licenced Date)	**Unlicenced**

If the SRF workflow is deleted, the status goes back to Requested to Licence (or Requested to Unlicence for removals).

Install Status (on the Equipment record) is driven by the SAF

When a SAF's actions are completed:

- Installed equipment is set to **Installed**, with the Installed Date.
- Removed equipment is set to **Removed**, with the Removed Date.

Both dates are the SAF's Works End Date, or today's date if it isn't set. The Equipment Action is also set to Complete.

Where each kind of change shows up

- **Equipment record:** this is where the current state of the equipment lives. Inventory, Site Info and SAF read it from here.
- **SRF Item:** stays as the historical record of what was requested. It isn't updated after approval or after installation.
- **SDS Equipment:** a snapshot of the Equipment record, taken when the SDS is created. Later changes to the Equipment record are **not** copied to existing SDS lines. On the SDS itself, staff can change Include, Install Status, Licence Status and Description, and those changes don't affect the Equipment record.
- **A later SRF that removes or swaps the equipment** copies the *current* Equipment values into its new SRF Item.

What this means for editing Equipment directly

- Editing descriptive fields (make, model, dimensions, height, azimuth, ...) while the related SRF is Submitted, Under Review, Feedback Issued or Resubmitted will be **overwritten** the next time the SRF is saved or re-submitted.
- Editing after that point will stick on the Equipment record. However, the change **won't** appear on the original SRF Item or on an SDS that already exists.
- Changing **Licence Status**, **Install Status**, the dates, **Parent TME**, **Associated Equipment Action**, site or customer by hand will put the equipment out of step with its SRF, SAF and Equipment Action. Later workflow steps may then overwrite the change or behave unexpectedly.

> ⚠️ **Disclaimer**
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> Please don't update data manually (Equipment, SRF Items, Equipment Actions, SDS Equipment or any related records), whether through the NetSuite UI, CSV import or scripts, without consulting us first. These records are linked and kept in sync by the application logic, which relies on the relationships and statuses described above. Manual changes can break the code and leave the data inconsistent ("corrupted") in ways that can be hard to detect and fix. If a change is needed, please contact us and we'll advise on the safest way to make it, or make it for you.